Why the Current Fixed-Lane ORAM Stash Depends on N

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Abstract

Circuit ORAM has an N-uniform, exponentially decaying snapshot stash tail in its proved parameter regime. The current fixed-lane ORAM transition does not exhibit this behavior. For a binary tree of depth $L = \log_2 N$, experiments and a product-state flow calculation indicate that one saturated lane carries only $\Theta(L^{-1/2})$ useful load. Consequently, maintaining a fixed operating margin requires approximately $Z = \Omega(\sqrt{\log_2 N})$ lanes. This note gives exact conservation identities, a mean-field explanation of the square-root law, empirical validation, and a comparison with Circuit ORAM. The conservation identities are exact; the stationary square-root law is a strongly supported conjecture, not yet a security theorem.

1 Setting

Consider the current binary fixed-lane transition with $N = 2^L$ logical blocks and Z physical lanes per bucket. Every access reads the path given by the requested block's old uniformly random label, removes the block, assigns it a fresh independent label, inserts it into the stash, admits stash blocks into free root lanes, and performs one lane-local eviction on the old path.

The physical paths are IID uniform under a fixed repeated-permutation logical schedule. The observed N-dependence is therefore not caused by biased or insufficiently random path selection. It is caused by the amount of useful flow that the fixed-lane transition can sustain through a depth-L routing network.

2 Exact conservation laws

Let S_t be post-writeback stash occupancy after operation t. During the next operation, let I_t equal one if the requested block is removed from the tree and zero if it was already in the stash. Let A_t be the number of stash blocks admitted into free root lanes. Block conservation gives the following identity without any probabilistic assumptions.

Proposition 1 (Root conservation)**.** *For every operation,*

$$S_{t+1} - S_t = I_t - A_t.$$

In stationarity,

$$\mathbb{E}[A_t] = \Pr[I_t = 1].$$

There is an analogous identity across every tree cut. Fix a depth d, and let $U_{d,t}$ be the number of blocks stored at depths $0, \ldots, d$. Let $R_{d,t}$ indicate removal of the requested block from this upper region, and let $C_{d,t}$ count blocks moved downward across the cut below depth d. Lane eviction never moves blocks upward, so

$$U_{d,t+1} - U_{d,t} = A_t - R_{d,t} - C_{d,t}.$$

Combining stationary balance with Proposition 1 yields

Proposition 2 (Cut conservation)**.** *In stationarity,*

$$\mathbb{E}[C_{d,t}] = \Pr[\textit{the requested block is in the tree below depth } d].$$

The upper d levels have only $Z(2^{d+1} - 1)$ slots. Hence a small-stash ORAM containing N blocks must place almost all blocks below every fixed shallow cut as N grows. Proposition 2 then requires nearly one aggregate downward crossing per operation through almost every shallow cut. Nominal tree capacity by itself is insufficient; the eviction process must provide the required flow at every depth.

3 Why lower occupancy does not create first-order slack

Suppose a small-stash system divides one unit of aggregate flow among Z lanes. Each lane must carry approximately

$$q \simeq \frac{1}{Z}$$

blocks per operation. At binary depth d, a block sees the correct next-bit eviction path once per 2^{d+1} global operations, while arrivals are split among 2^d nodes. In dilute flow, Little's-law reasoning therefore gives a typical lane-slot occupancy

$$\rho \simeq 2q.$$

Thus adding lanes really does lower occupancy. However, a useful move also requires an occupied source. The ideal binary flow from one lane is $\rho/2 \simeq q$, exactly the offered load. The first-order benefit of more holes is cancelled by the first-order loss of occupied sources, leaving no constant service margin.

For an independent Bernoulli path state with occupied probability ρ, a direct expansion of the production mask's local cut current gives

$$J_2(\rho) = \frac{\rho}{2} - \frac{3}{64}\rho^3 + O(\rho^4). \tag{1}$$

The quadratic term cancels because the mask can jump over one incompatible occupied blocker. Substituting $\rho \simeq 2q$ gives

$$J_2(2q) = q - \frac{3}{8}q^3 + O(q^4). \tag{2}$$

Consequently, one routing level loses a relative $O(q^2)$ fraction of ideal flow.

A lane storing a non-negligible fraction of N blocks must route blocks down to depth $L - O(\log L)$, so the loss is accumulated over $\Theta(L)$ prefix stages. Equations (1)–(2) therefore suggest that the relevant dimensionless congestion parameter is

$$Lq^2.$$

Nonvanishing end-to-end flow requires $Lq^2 = O(1)$, giving

$$q_L = \Theta(L^{-1/2}). \tag{3}$$

This is a mean-field scaling argument rather than a proof for the correlated stationary ORAM process.

An equivalent intuition views a vacancy created deep in a lane as prefix credit that must propagate back to the root. Random left/right demand and eviction opportunities produce an approximately zero-drift reflected walk. Its typical imbalance or maximum over L stages is $\Theta(\sqrt{L})$, so the usable load is its reciprocal. Fixed lane identity prevents imbalances and vacancies from being pooled across lanes.

4 Measured per-lane law

Production-equivalent saturated one-lane experiments used direct operating system randomness, random initial labels, fresh independent labels on every access, and at least $64N$ warm-up operations. Define

$$q_L = \frac{\mathbb{E}[\text{blocks resident in one lane tree}]}{N}.$$

The measured products were

L	$q_L\sqrt{L}$
10	0.6317
12	0.6221
14	0.6222
16	0.6246
18	0.6317
20	0.6327

These results support

$$q_L \simeq \frac{c}{\sqrt{L}}, \qquad c \simeq 0.627. \tag{4}$$

The fitted constant is strikingly close to $\sqrt{\pi/8} = 0.626657\ldots$. A random-walk maximum heuristic can produce this value, but the required mapping and its factor of two have not been proved. The constant must therefore be regarded as empirical.

When the global stash is backlogged, the Z marginal lanes behave approximately as parallel saturated copies. This gives the fluid approximation

$$\boxed{\frac{\mathbb{E}[S]}{N} \simeq \left(1 - \frac{0.627Z}{\sqrt{\log_2 N}}\right)_+.} \tag{5}$$

The corresponding crossover height is

$$L_{\text{crit}}(Z) \simeq (0.627Z)^2, \qquad Z_{\text{crit}}(L) \simeq 1.595\sqrt{L}. \tag{6}$$

Z	Predicted L_{crit}	Observed behavior
5	9.8	degradation near $N = 2^{10}$
6	14.2	degradation near $N = 2^{14}$
7	19.3	small through 2^{18}, large at 2^{20}
8	25.2	small but increasingly heavy-tailed through 2^{20}
9	31.8	small through 2^{20}; no asymptotic guarantee

The abrupt transition is ordinary queue stability. For example, at $Z = 7$ the fitted aggregate capacity is 1.034 at $L = 18$ but 0.981 at $L = 20$. A few percent change moves the queue from slightly negative to slightly positive drift. Since $N = 2^L$, adding one lane can postpone the transition by a very large multiplicative factor in N.

5 Direct $Z = 6$ and $Z = 7$ results

The current block-conserving simulator produced the following mean post-writeback stash occupancies.

N	$Z = 6$	$Z = 7$
2^{10}	0.773	0.076
2^{12}	6.288	0.222
2^{14}	207.214	0.762
2^{16}	4314.083	3.668
2^{18}	29660.358	32–41
2^{20}	—	9416.649

At $N = 2^{20}$, the $Z = 7$ measured range was 1542–16682 after a 2^{26}-operation warm-up. Thus its excellent small-N behavior was a finite-height effect, not an N-uniform stash bound.

At $N = 2^{20}$, current $Z = 8$ had mean stash 0.873 but transient insertion demand exceeded 20 with probability approximately 1.10%. Current $Z = 9$ had mean 0.049 and exceedance probability approximately 1.33×10^{-4}. These finite-run probabilities are snapshot estimates, not lifetime failure bounds or cryptographic extrapolations.

6 Greedy fixed-lane control

A corrected, conserving deepest-hole transition improves the per-lane constant but does not change the observed exponent:

$$q_L^{\text{greedy}} \simeq \frac{0.82}{\sqrt{L}}.$$

At $N = 2^{20}$, greedy $Z = 6$ and $Z = 7$ had mean post-stash occupancies 2.529 and 0.095, respectively, compared with 9416.649 for current $Z = 7$. The predicted greedy crossover heights are approximately 24 for $Z = 6$ and 33 for $Z = 7$. Greedy movement therefore matters substantially, but the fixed-lane routing evidence still suggests $Z = \Omega(\sqrt{\log_2 N})$ for binary trees.

7 Why Circuit ORAM is different

Circuit ORAM's proved snapshot stash bound has the form

$$\Pr[S > R] \leq C\alpha^R, \qquad \alpha < 1,$$

with constants independent of N in the theorem's parameter regime. Therefore its expected snapshot stash is $O(1)$ independently of N; only a separate lifetime union bound forces R to grow with the number of operations or desired security parameter.

Circuit ORAM does not route vacancies through permanently separated channels. It pools bucket slots, chooses the deepest compatible candidate from the stash and path, moves directly into deep vacancies, and performs two eviction paths per logical access. These mechanisms provide an N-uniform service margin. Lane ORAM does have Z potential parallel chains, but the useful capacity of each fixed chain decreases with routing depth because small local losses accumulate at every prefix stage.

8 General branching factors

Binary branching is unusually favorable. In the same independent path-state model, general branching factor B gives

$$J_B(\rho) = \frac{\rho}{B} - \frac{(B-1)(B-2)}{B^3}\rho^2 + O(\rho^3).$$

The quadratic loss cancels only for $B = 2$. Since dilute occupancy satisfies $\rho \simeq Bq$, for $B > 2$ the relative loss per level is $O(q)$ rather than $O(q^2)$. Across $L = \log_B N$ levels this suggests

$$q_L = \Theta(1/L), \qquad Z = \Omega(\log_B N).$$

Existing $B = 4$ data fit $q_L L \simeq 0.363$, and independent Z values infer the same marginal per-lane capacity. This explains why the tested $B = 4$ and $B = 8$ configurations perform worse than their raw number of simultaneous lane moves suggests.

9 Status of the analysis

Propositions 1 and 2 are exact consequences of block conservation. Equation (1) is exact for an IID Bernoulli product-state model of a loaded path. The extension from that local model to the correlated stationary ORAM process, the square-root stationary law, and the numerical constant in Equation (4) remain conjectural. Their unusually accurate agreement across tree sizes and lane counts makes them a strong explanation of the observed behavior, but not a replacement for a formal stash-security proof.